Determining the viscosity of a colloidal solution of monodisperse polystyrene microspheres in water by tracking brownian motion

Bela F. Brunner[1] and Adam Zich[1]
[1]*Department of Physics, Utrecht University*
(Dated: October 30, 2026)

Abstract

I. INTRODUCTION

Brownian motion is a phenomenon where small particles in a colloidal solution are not static, but are "jiggling about". This phenomenon is named after Robert Brown, who observed it on pollen grains in the 19th century. The theory of it has then been developed further by Christian Wiener and Albert Einstein. [1] It is known that the amplitude of the motion is larger, the smaller the particles are, the lower the viscosity of the fluid and the higher the temperature of the solution. [2] An exact formula for the mean displacement of the particles has been found by Einstein in 1905. [3] In this article we look at determining the viscosity of a colloidal solution of monodisperse polystyrene microspheres in water.

II. THEORY

theory

III. METHODOLOGY

Methods As samples to be analysed, we are using monodisperse polystyrene microspheres of the sizes of $0.5 - 2\mu$m who are suspensed in water [4]. The resulting liquid is a collodial suspension, which is the

IV. RESULTS

Results

V. DISCUSSION

Discussion

VI. CONCLUSIONS

Conclusion

[1] Stephen J. Blundell and Katherine M. Blundell. *Concepts in Thermal Physics*. Oxford University Press, Oxford, 2nd edition, 2009.
[2] The Editors of Encyclopaedia Britannica. Brownian motion. Encyclopædia Britannica, 2026. Accessed: 2026-10-30.
[3] Albert Einstein. Über die von der molekularkinetischen theorie der wärme geforderte bewegung von in ruhenden flüssigkeiten suspendierten teilchen. *Annalen der Physik*, 322(8):549–560, 1905.
[4] Polybead® Microspheres 2.00µm — Polysciences, Inc.